Table S9 Joint-ranking composition and directional concordance

A1. Method-specific top-10% counts

Target pair	Scheme	N	k	Dual in population	Dual selected	A-only selected	B-only selected
EGFR/HER2	S1	97	10	28	1	4	5
EGFR/HER2	S2	97	10	28	2	1	7
EGFR/HER2	S3	97	10	28	2	4	4
EGFR/HER2	S4	97	10	28	2	2	6
EGFR/HER2	S5	97	10	28	2	0	8
JAK1/JAK2	S1	96	10	32	6	3	1
JAK1/JAK2	S2	96	10	32	7	3	0
JAK1/JAK2	S3	96	10	32	4	4	2
JAK1/JAK2	S4	96	10	32	4	3	3
JAK1/JAK2	S5	96	10	32	7	3	0
JAK1/TYK2	S1	96	10	32	3	1	6
JAK1/TYK2	S2	96	10	32	6	2	2
JAK1/TYK2	S3	96	10	32	5	0	5
JAK1/TYK2	S4	96	10	32	5	3	2
JAK1/TYK2	S5	96	10	32	5	4	1
PIK3CA/mTOR	S1	96	10	32	4	0	6
PIK3CA/mTOR	S2	96	10	32	6	3	1
PIK3CA/mTOR	S3	96	10	32	7	1	2
PIK3CA/mTOR	S4	96	10	32	5	3	2
PIK3CA/mTOR	S5	96	10	32	7	1	2
AChE/BChE	S1	80	8	27	6	0	2
AChE/BChE	S2	79	8	27	3	2	3
AChE/BChE	S3	80	8	27	3	3	2
AChE/BChE	S4	79	8	27	3	3	2
AChE/BChE	S5	82	9	27	3	1	5
F2/F10	S1	96	10	32	4	4	2
F2/F10	S2	95	10	31	4	3	3
F2/F10	S3	96	10	32	7	1	2
F2/F10	S4	95	10	31	5	5	0
F2/F10	S5	95	10	31	6	3	1

Panel A uses each method's own members and is descriptive. Table 2 and panel B use separate matched populations for every comparison. Joint rank = max(rank A, rank B), with average single-target ranks; ties are resolved by rank sum then global molecular ID. Both differences are new scheme minus S1. Opposite signs define concordant improvement or worsening; equal nonzero signs are discordant; either exact zero defines no change. S4-S1 is supporting, using verified identical directional members and no additional CI. Complete paired composition, per-ligand ranks and all CAP5 ranking results are supplied as data files.

Table S9 Joint-ranking composition and directional concordance (continued)

A1. Method-specific top-10% counts

Target pair	Scheme	N	k	Dual in population	Dual selected	A-only selected	B-only selected
PPARG/PPARA	S1	96	10	32	6	3	1
PPARG/PPARA	S2	94	10	30	2	4	4
PPARG/PPARA	S3	96	10	32	4	3	3
PPARG/PPARA	S4	94	10	30	3	4	3
PPARG/PPARA	S5	96	10	32	1	4	5
PPARA/PPARD	S1	96	10	32	3	2	5
PPARA/PPARD	S2	91	10	27	5	2	3
PPARA/PPARD	S3	96	10	32	6	2	2
PPARA/PPARD	S4	91	10	27	4	1	5
PPARA/PPARD	S5	96	10	32	5	1	4

Panel A uses each method's own members and is descriptive. Table 2 and panel B use separate matched populations for every comparison. Joint rank = max(rank A, rank B), with average single-target ranks; ties are resolved by rank sum then global molecular ID. Both differences are new scheme minus S1. Opposite signs define concordant improvement or worsening; equal nonzero signs are discordant; either exact zero defines no change. S4-S1 is supporting, using verified identical directional members and no additional CI. Complete paired composition, per-ligand ranks and all CAP5 ranking results are supplied as data files.

Table S9 Joint-ranking composition and directional concordance

A2. Method-specific top-10% fractions

Target pair	Scheme	Dual fraction	Single-target fraction	Dual recall	Population dual fraction
EGFR/HER2	S1	0.1000	0.9000	0.0357	0.2887
EGFR/HER2	S2	0.2000	0.8000	0.0714	0.2887
EGFR/HER2	S3	0.2000	0.8000	0.0714	0.2887
EGFR/HER2	S4	0.2000	0.8000	0.0714	0.2887
EGFR/HER2	S5	0.2000	0.8000	0.0714	0.2887
JAK1/JAK2	S1	0.6000	0.4000	0.1875	0.3333
JAK1/JAK2	S2	0.7000	0.3000	0.2188	0.3333
JAK1/JAK2	S3	0.4000	0.6000	0.1250	0.3333
JAK1/JAK2	S4	0.4000	0.6000	0.1250	0.3333
JAK1/JAK2	S5	0.7000	0.3000	0.2188	0.3333
JAK1/TYK2	S1	0.3000	0.7000	0.0938	0.3333
JAK1/TYK2	S2	0.6000	0.4000	0.1875	0.3333
JAK1/TYK2	S3	0.5000	0.5000	0.1562	0.3333
JAK1/TYK2	S4	0.5000	0.5000	0.1562	0.3333
JAK1/TYK2	S5	0.5000	0.5000	0.1562	0.3333
PIK3CA/mTOR	S1	0.4000	0.6000	0.1250	0.3333
PIK3CA/mTOR	S2	0.6000	0.4000	0.1875	0.3333
PIK3CA/mTOR	S3	0.7000	0.3000	0.2188	0.3333
PIK3CA/mTOR	S4	0.5000	0.5000	0.1562	0.3333
PIK3CA/mTOR	S5	0.7000	0.3000	0.2188	0.3333
AChE/BChE	S1	0.7500	0.2500	0.2222	0.3375
AChE/BChE	S2	0.3750	0.6250	0.1111	0.3418
AChE/BChE	S3	0.3750	0.6250	0.1111	0.3375
AChE/BChE	S4	0.3750	0.6250	0.1111	0.3418
AChE/BChE	S5	0.3333	0.6667	0.1111	0.3293
F2/F10	S1	0.4000	0.6000	0.1250	0.3333
F2/F10	S2	0.4000	0.6000	0.1290	0.3263
F2/F10	S3	0.7000	0.3000	0.2188	0.3333
F2/F10	S4	0.5000	0.5000	0.1613	0.3263
F2/F10	S5	0.6000	0.4000	0.1935	0.3263

Panel A uses each method's own members and is descriptive. Table 2 and panel B use separate matched populations for every comparison. Joint rank = max(rank A, rank B), with average single-target ranks; ties are resolved by rank sum then global molecular ID. Both differences are new scheme minus S1. Opposite signs define concordant improvement or worsening; equal nonzero signs are discordant; either exact zero defines no change. S4-S1 is supporting, using verified identical directional members and no additional CI. Complete paired composition, per-ligand ranks and all CAP5 ranking results are supplied as data files.

Table S9 Joint-ranking composition and directional concordance (continued)

A2. Method-specific top-10% fractions

Target pair	Scheme	Dual fraction	Single-target fraction	Dual recall	Population dual fraction
PPARG/PPARA	S1	0.6000	0.4000	0.1875	0.3333
PPARG/PPARA	S2	0.2000	0.8000	0.0667	0.3191
PPARG/PPARA	S3	0.4000	0.6000	0.1250	0.3333
PPARG/PPARA	S4	0.3000	0.7000	0.1000	0.3191
PPARG/PPARA	S5	0.1000	0.9000	0.0312	0.3333
PPARA/PPARD	S1	0.3000	0.7000	0.0938	0.3333
PPARA/PPARD	S2	0.5000	0.5000	0.1852	0.2967
PPARA/PPARD	S3	0.6000	0.4000	0.1875	0.3333
PPARA/PPARD	S4	0.4000	0.6000	0.1481	0.2967
PPARA/PPARD	S5	0.5000	0.5000	0.1562	0.3333

Panel A uses each method's own members and is descriptive. Table 2 and panel B use separate matched populations for every comparison. Joint rank = max(rank A, rank B), with average single-target ranks; ties are resolved by rank sum then global molecular ID. Both differences are new scheme minus S1. Opposite signs define concordant improvement or worsening; equal nonzero signs are discordant; either exact zero defines no change. S4-S1 is supporting, using verified identical directional members and no additional CI. Complete paired composition, per-ligand ranks and all CAP5 ranking results are supplied as data files.

Table S9 Joint-ranking composition and directional concordance

B. Paired changes versus Vina

Target pair	Scheme vs S1	Ranking common n	Directional common dual n	Minimum AUROC difference	Single-target fraction difference	Concordance	Supporting comparison
EGFR/HER2	S2	97	28	-0.0257	-0.1000	Discordant	NO
EGFR/HER2	S3	97	28	0.0167	-0.1000	Concordant improvement	NO
EGFR/HER2	S4	97	28	0.0346	-0.1000	Concordant improvement	YES
EGFR/HER2	S5	97	28	-0.0011	-0.1000	Discordant	NO
JAK1/JAK2	S2	96	32	-0.1523	-0.1000	Discordant	NO
JAK1/JAK2	S3	96	32	-0.1172	0.2000	Concordant worsening	NO
JAK1/JAK2	S4	96	32	-0.1465	0.2000	Concordant worsening	YES
JAK1/JAK2	S5	96	32	-0.0771	-0.1000	Discordant	NO
JAK1/TYK2	S2	96	32	0.1377	-0.3000	Concordant improvement	NO
JAK1/TYK2	S3	96	32	0.0654	-0.2000	Concordant improvement	NO
JAK1/TYK2	S4	96	32	0.2539	-0.2000	Concordant improvement	YES
JAK1/TYK2	S5	96	32	0.2471	-0.2000	Concordant improvement	NO
PIK3CA/mTOR	S2	96	32	-0.0049	-0.2000	Discordant	NO
PIK3CA/mTOR	S3	96	32	0.1240	-0.3000	Concordant improvement	NO
PIK3CA/mTOR	S4	96	32	-0.0635	-0.1000	Discordant	YES
PIK3CA/mTOR	S5	96	32	0.0176	-0.3000	Concordant improvement	NO
AChE/BChE	S2	79	27	-0.1416	0.3750	Concordant worsening	NO
AChE/BChE	S3	80	27	-0.1433	0.3750	Concordant worsening	NO
AChE/BChE	S4	79	27	-0.2431	0.3750	Concordant worsening	YES
AChE/BChE	S5	80	27	-0.1830	0.3750	Concordant worsening	NO

Panel A uses each method's own members and is descriptive. Table 2 and panel B use separate matched populations for every comparison. Joint rank = max(rank A, rank B), with average single-target ranks; ties are resolved by rank sum then global molecular ID. Both differences are new scheme minus S1. Opposite signs define concordant improvement or worsening; equal nonzero signs are discordant; either exact zero defines no change. S4-S1 is supporting, using verified identical directional members and no additional CI. Complete paired composition, per-ligand ranks and all CAP5 ranking results are supplied as data files.

Table S9 Joint-ranking composition and directional concordance (continued)

B. Paired changes versus Vina

Target pair	Scheme vs S1	Ranking common n	Directional common dual n	Minimum AUROC difference	Single-target fraction difference	Concordance	Supporting comparison
F2/F10	S2	95	31	0.1643	0	No change	NO
F2/F10	S3	96	32	0.2471	-0.3000	Concordant improvement	NO
F2/F10	S4	95	31	0.1956	-0.1000	Concordant improvement	YES
F2/F10	S5	95	31	0.1815	-0.2000	Concordant improvement	NO
PPARG/PPARA	S2	94	30	-0.3089	0.4000	Concordant worsening	NO
PPARG/PPARA	S3	96	32	-0.2915	0.2000	Concordant worsening	NO
PPARG/PPARA	S4	94	30	-0.2578	0.3000	Concordant worsening	YES
PPARG/PPARA	S5	96	32	-0.1558	0.5000	Concordant worsening	NO
PPARA/PPARD	S2	91	27	0.1192	-0.2000	Concordant improvement	NO
PPARA/PPARD	S3	96	32	0.1582	-0.3000	Concordant improvement	NO
PPARA/PPARD	S4	91	27	0	-0.1000	No change	YES
PPARA/PPARD	S5	96	32	0.1230	-0.2000	Concordant improvement	NO

Panel A uses each method's own members and is descriptive. Table 2 and panel B use separate matched populations for every comparison. Joint rank = max(rank A, rank B), with average single-target ranks; ties are resolved by rank sum then global molecular ID. Both differences are new scheme minus S1. Opposite signs define concordant improvement or worsening; equal nonzero signs are discordant; either exact zero defines no change. S4-S1 is supporting, using verified identical directional members and no additional CI. Complete paired composition, per-ligand ranks and all CAP5 ranking results are supplied as data files.

Table S9 Joint-ranking composition and directional concordance

C. Concordance counts

Scheme vs S1	Pairs	Concordant improvement	Concordant worsening	Discordant	No change
S2	8	2	2	3	1
S3	8	5	3	0	0
S4	8	3	3	1	1
S5	8	4	2	2	0

Panel A uses each method's own members and is descriptive. Table 2 and panel B use separate matched populations for every comparison. Joint rank = max(rank A, rank B), with average single-target ranks; ties are resolved by rank sum then global molecular ID. Both differences are new scheme minus S1. Opposite signs define concordant improvement or worsening; equal nonzero signs are discordant; either exact zero defines no change. S4-S1 is supporting, using verified identical directional members and no additional CI. Complete paired composition, per-ligand ranks and all CAP5 ranking results are supplied as data files.